

Participant: NUCLEAR

You are responsible for the **NUCLEAR** technology in the market.

In real life, a bid includes multiple energy and price segments. However, for simplicity's sake, in this game you will only offer one segment. Therefore, your bid for a specific round will consist of the following:

- The **price** per megawatt-hour (€/MWh) at which you are willing to sell your energy. The minimum value (floor) is 0 €/MWh. The maximum value (cap) is 180 €/MWh.
- The amount of **energy** you are offering (GWh). The minimum value is 0 GWh. The maximum is 8 GWh.

Take into consideration that:

1. Your unitary variable cost associated with fuel consumption is **5 €/MWh**.
2. You don't have CO₂ emissions associated with generation.
3. Nuclear plants must work at full capacity. This means that, regardless the energy you sell in each round, your plant will deliver 10 GWh. If you sell less than that, you will pay a penalty (because you will inject more energy than that compromised in the market, thus you create an imbalance). The penalty price is different for each round, and depends on the overall imbalance, considering deviations from all participants.

For example, if you sell 6 GWh in the market, since you actually always generate 8 GWh, you create an imbalance of 2 GWh. If the market clearing price is 20 €/MWh:

- You earn $6 \times 20 = 120$ thousand € (market incomes).
- You pay $8 \times 5 = 40$ thousand € for the fuel consumption (variable cost).
- If the penalty price were 30 €/MWh, you would pay a penalty of $2 \times 30 = 60$ thousand €.
- Thus, overall, you earn $120 - 40 - 60 = 20$ thousand € (total incomes).